

AI Gym & Fitness Assistant

Project Report

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Chapter 1: Introduction

Health and fitness have become an important part of daily life. Many people struggle to maintain proper workout schedules and healthy diets. The AI Gym & Fitness Assistant is a web-based application that helps users monitor their health, calculate BMI, receive workout and diet recommendations, track water intake, estimate calorie requirements, and interact with an AI fitness chatbot.

The project has been developed using Python, Streamlit, and SQLite to provide a simple, user-friendly interface.

Chapter 2: Objectives

The objectives of this project are:

- Develop a user-friendly fitness assistant.
- Calculate BMI accurately.
- Recommend workouts based on BMI.
- Suggest healthy diet plans.
- Track daily water intake.
- Calculate daily calorie requirements.
- Maintain workout history.
- Provide an AI-based fitness chatbot.
- Store user data securely in SQLite.

Chapter 3: Problem Statement

Many individuals find it difficult to maintain a consistent fitness routine because they lack personalized guidance. Existing applications are often expensive or provide too many unnecessary features. This project aims to provide an easy-to-use and affordable fitness assistant with personalized recommendations.

Chapter 4: Proposed Solution

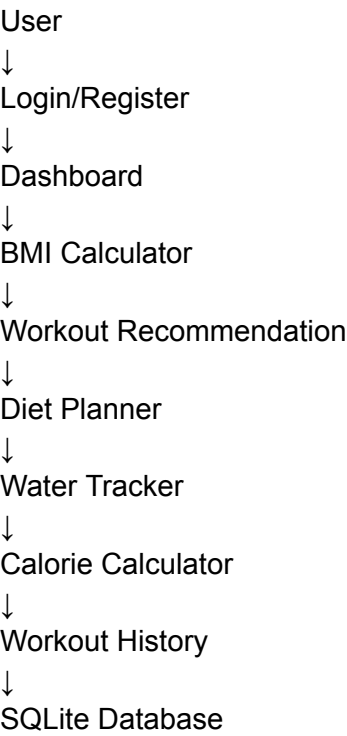
The proposed system provides:

- User Registration
- Secure Login
- BMI Calculator
- Workout Recommendation
- Diet Recommendation
- Water Intake Tracker
- Calorie Calculator
- Workout History
- Dashboard
- AI Chatbot

Chapter 5: Technologies Used

Technology	Purpose
Python	Programming Language
Streamlit	Web Application Framework
SQLite	Database
Pandas	Data Handling
Google Gemini API	AI Chatbot
VS Code	IDE

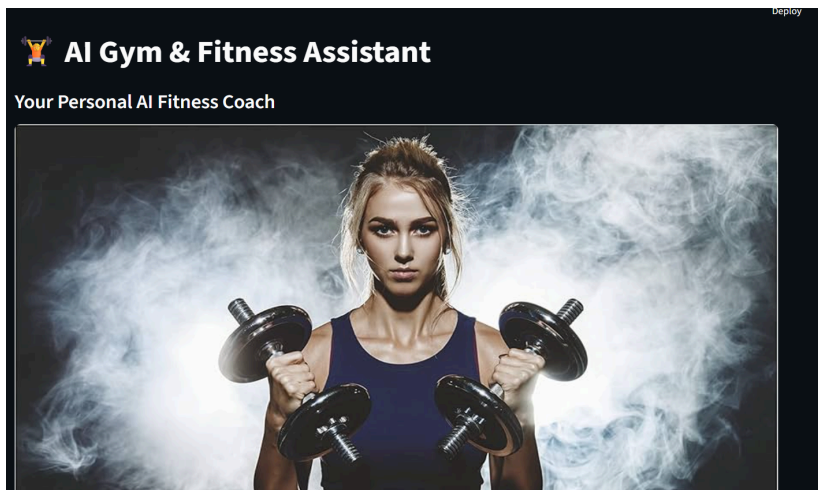
Chapter 6: System Architecture



Chapter 7: Project Modules

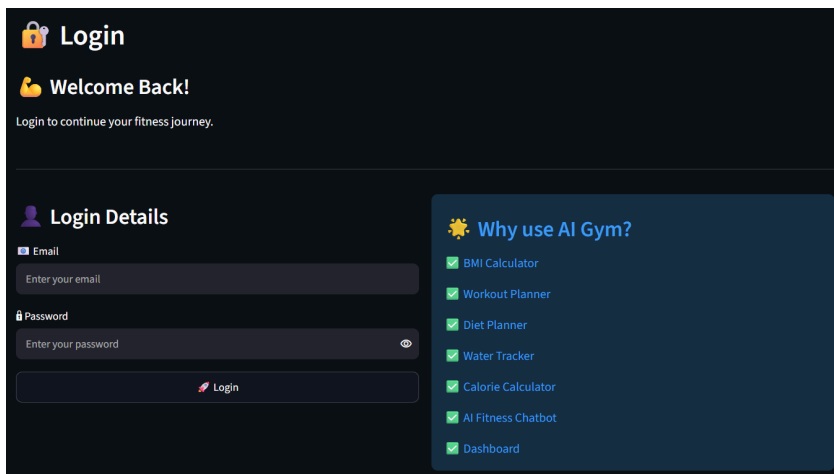
1. Home Page

Provides navigation to all modules.



2. Login Module

Allows registered users to securely log in.

The image displays the login interface of the AI Gym & Fitness Assistant. It features a dark blue background. At the top, there is a 'Login' header with a lock icon, followed by a 'Welcome Back!' message and a sub-header 'Login to continue your fitness journey.'. Below this, there is a 'Login Details' section with two input fields: 'Email' and 'Password'. The 'Email' field has a placeholder 'Enter your email' and the 'Password' field has a placeholder 'Enter your password' and a toggle icon. A 'Login' button is positioned below the password field. To the right of the login details, there is a section titled 'Why use AI Gym?' with a list of features, each preceded by a green checkmark: 'BMI Calculator', 'Workout Planner', 'Diet Planner', 'Water Tracker', 'Calorie Calculator', 'AI Fitness Chatbot', and 'Dashboard'.

3. Registration Module

Stores user information in SQLite.



User Registration

Enter your Name

Enter your Email

Enter your Password

Enter your Age

Enter your Height (in meters)

Enter your Weight (in kg)

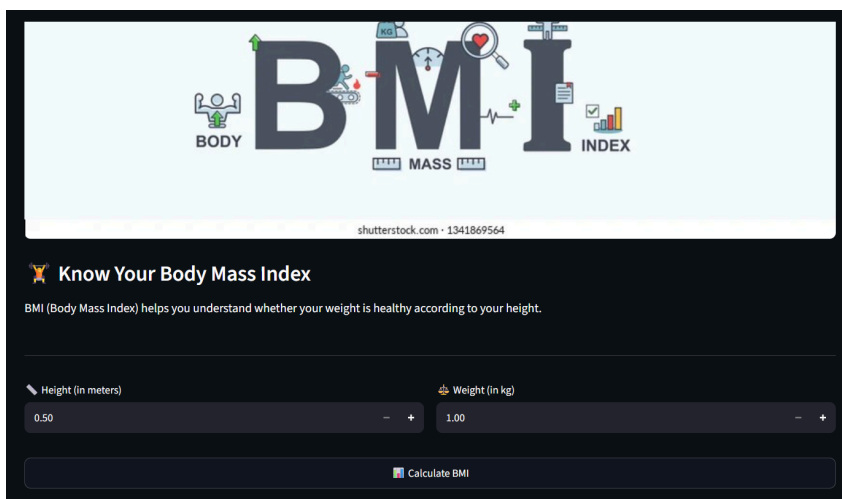
Register

4. BMI Calculator

Calculates BMI using height and weight.

Displays:

- Underweight
- Normal
- Overweight
- Obese



Know Your Body Mass Index

BMI (Body Mass Index) helps you understand whether your weight is healthy according to your height.

Height (in meters) Weight (in kg)

0.50 1.00

Calculate BMI

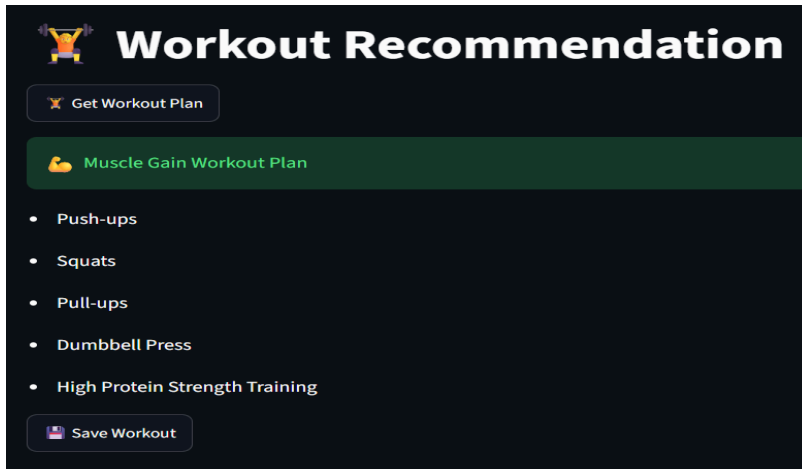
5. Workout Recommendation

Suggests workouts according to BMI.

Examples:

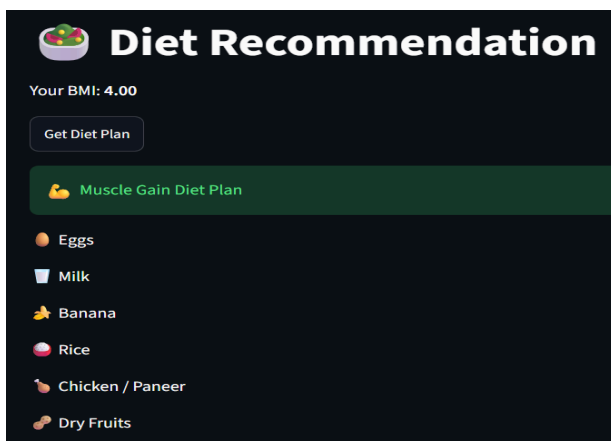
- Muscle Gain
- Balanced Fitness

- Fat Loss
- Obesity Workout



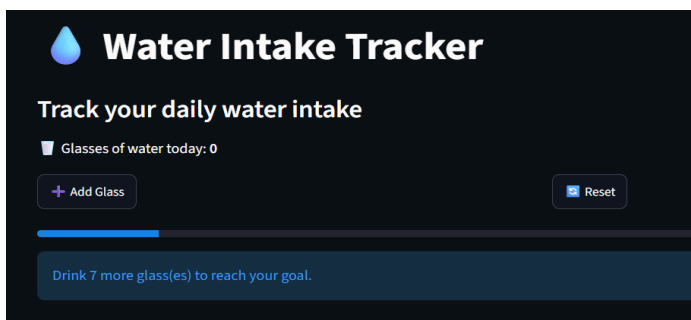
6. Diet Planner

Provides diet recommendations according to BMI.



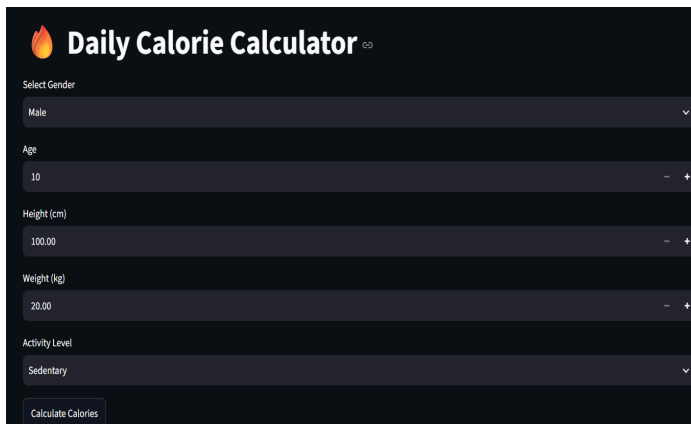
7. Water Tracker

Tracks daily water intake.



8. Calorie Calculator

Estimates daily calorie requirements.

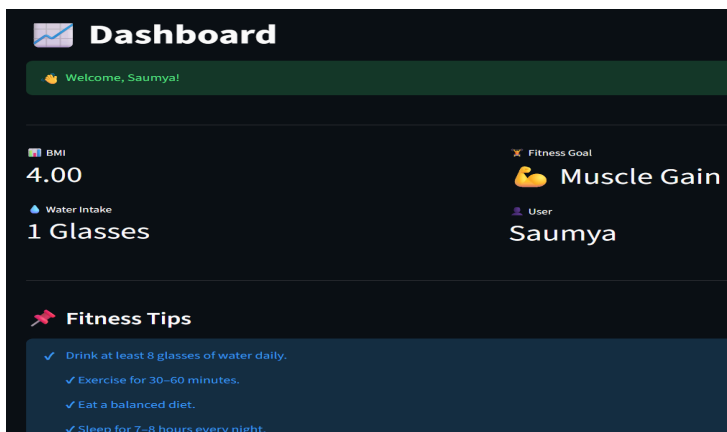


The image shows a web interface for a "Daily Calorie Calculator". It features a dark theme with a light blue header. The form includes fields for "Select Gender" (a dropdown menu with "Male" selected), "Age" (a numeric input field with "10"), "Height (cm)" (a numeric input field with "100.00"), "Weight (kg)" (a numeric input field with "20.00"), and "Activity Level" (a dropdown menu with "Sedentary" selected). A "Calculate Calories" button is located at the bottom left of the form.

9. Dashboard

Displays:

- Username
- BMI
- Water Intake
- Fitness Goal



The image shows a web interface for a "Dashboard". It features a dark theme with a light blue header. The dashboard includes a "Welcome, Saumya!" message, a "BMI" section showing "4.00", a "Water Intake" section showing "1 Glasses", a "Fitness Goal" section showing "Muscle Gain", and a "User" section showing "Saumya". A "Fitness Tips" section at the bottom lists four tips: "Drink at least 8 glasses of water daily.", "Exercise for 30-60 minutes.", "Eat a balanced diet.", and "Sleep for 7-8 hours every night."

10. Workout History

Stores workout history in SQLite and displays previous workout records.

Workout History	
Your Saved Workouts	
workout	date
Muscle Gain Workout	2026-07-26 10:50:06

11. AI Chatbot

Provides basic fitness-related guidance using AI.



AI Fitness Assistant

Ask your fitness question

Ask AI

Chapter 8: Database Design

Users Table

Field	Type
id	Integer
name	Text
email	Text
password	Text
age	Integer
height	Real
weight	Real

Workout History Table

Field	Type
id	Integer
username	Text
e	
workout	Text
date	Timestamp

Chapter 9: Project Workflow

Start



Register



Login



Dashboard



Calculate BMI



Workout Recommendation



Diet Recommendation



Water Tracking



Calorie Calculation



Workout History



Logout

Chapter 10: Testing

Test Case	Expected Result	Status
Registration	User Registered	Pass
Login	Successful Login	Pass
BMI Calculator	Correct BMI	Pass
Workout Recommendation	Recommendation Displayed	Pass
Diet Recommendation	Recommendation Displayed	Pass
Water Tracker	Water Updated	Pass
Calorie Calculator	Calories Calculated	Pass
Workout History	Workout Saved	Pass
Dashboard	Data Displayed	Pass

Chapter 11: Results

The developed application successfully performs:

- User Authentication
- BMI Calculation
- Workout Recommendation
- Diet Recommendation
- Water Tracking
- Calorie Estimation
- Workout History Storage
- Dashboard Display
- AI Chatbot Interaction

The system provides a simple and user-friendly interface for fitness management.

Chapter 12: Future Scope

Future improvements include:

- AI-based pose detection using MediaPipe.
- Real-time workout repetition counting.
- Personalized ML-based workout recommendations.
- Cloud database integration.
- Mobile application development.
- Wearable device integration.
- Advanced analytics and progress charts.
- Admin dashboard for user analytics.

Chapter 13: Conclusion

The AI Gym & Fitness Assistant successfully demonstrates the integration of Artificial Intelligence concepts with modern web application development. It provides users with personalized health information, workout recommendations, and diet suggestions through an intuitive interface. The project enhances user awareness about health and fitness while showcasing practical applications of Python, Streamlit, SQLite, and AI technologies.